

Q. 1 (Turkeys). It is nearly Thanksgiving, when everyone will be cooking a turkey for the holiday meal. The supermarket is trying to meet demand. Frozen Turkeys are delivered to the supermarket according to a Poisson process with an average rate of 50 per day (the store is open 24/7). Each turkey's weight is exponentially distributed with mean 15 lbs, independently of each other and of the arrival times.

- a. A customer calls and places an order for 10 turkeys that weigh at least 10 lbs each, to be picked up exactly one day later. When the customer calls, the store has no turkeys in stock. What is the probability that exactly 10 turkeys that weigh at least 10 lbs arrived at the store by the time the customer comes to pick up the order? (You may assume these turkeys are set aside for the customer when they come in.)
- b. At the beginning of the next day, the store is out of turkeys again. Customers arrive at the store according to a Poisson process with an average rate of 10 per hour, independently of the turkeys. Each customer buys a turkey, provided a turkey is in stock when the customer arrives. What is the probability that the second customer to enter the store that day does *not* get a turkey?

Q. 2 (Pie). You bake a pumpkin pie for Thanksgiving from a sugar pumpkin you bought at the supermarket. The size of the pie you end up with depends on the size of the pumpkin you bought: the number of slices it serves is gamma distributed with shape 2 and scale $\frac{1}{5}$. (The number of slices does not need to be integer, e.g., you could have a pie that serves 12.7 slices.) Unfortunately, before you have a chance to share the pie with your friends, your dog finds it and eats a random fraction of the pie. The fraction eaten by your dog is uniformly distributed (in $[0, 1]$) and independent of the pie's size.

- a. What are the mean and variance of the number of slices of pie that are left?
- b. Given that your dog left exactly 4 slices, what is the expected size of the original pie (before your dog had his way with it)?

Q. 3 (Indigestion). The Thanksgiving meal is over, and you have grossly over-eaten. Every hour from now on, you will get either 0, 1, or 2 bouts of indigestion with equal probability. The number of bouts of indigestion in each hour is independent. At the end of each hour (but not during the hour), you consider whether you should take some peptobismol to relieve the symptoms. Peptobismol should only be taken after you have already suffered at least 3 bouts of indigestion.

- a. What is the probability that you will never have to take peptobismol?
- b. What is the expected number of hours after which you take peptobismol?